# Test Report 8572674.

## Captiv Fenestration / Prominance UPVC Profiles



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# Introduction.

This report has been prepared by Errol Creary and relates to the activity detailed below:

| Job/Registration Details | | Client Details |
|---|---|---|
| **Job number:** | **8572674** | Captiv Fenestration |
| Job type: | Testing Samples Submitted | Appanaikenpatti, Sulur |
| Start Date: | 25/07/2016 | Coimbatore |
| Test type: | Direct | Tamilnadu |
| Sample ID: | 10164761 | 641402 |
| | 10165196 | India |
| | 10173643 | |
| **Registration:** | **NA** | |
| Protocol: | NA | |
| Quality system: | NA | |
| **Registration:** | **NA** | |
| Protocol: | NA | |
| Quality system: | NA | |

The report has been approved for issue by Mark Manito – Team Manager

This issue supersedes all previous issues. The amendment giving rise to this issue of the Report can be ascertained by contacting the authorizing signatory.

| Approved For Issue | |
|---|---|
| M. Manito | Issue Date: 12 September 2017 |

# Objectives.

Direct test

# Product Scope.

PC 62-US-04

# Report Summary.

The samples were received on 25 July 2016 and the testing was started on 25 July 2016.

The samples submitted complied with the requirements of the test work conducted.

## Test Samples.

| Sample Id | ER Number | Description |
|---|---|---|
| 1 | PC 62-US-04 | Casement Inward Door Sash |

## Description of Test Samples.

| Sample Description |
|---|
| 1 off Main Profile(inc 1 off PAS 23 & PAS 24 door profile) |
| 4 off 250mm (+10/-10) lengths |
| 2 off 200mm (+10/-10) lengths |
| 12 off 300mm (+/-5mm) lengths |
| 1 off 1000mm (+10/-10) lengths |
| 10 off welded 90° corners with 500mm long legs "I" and "V" values for each welded profile |

# Test Requirements.

BS EN 12608 Direct

| Clause | Requirements | |
|---|---|---|
| 5. | Requirements | |
| 5.1 | Materials | |
| 5.1.3 | Material characteristics | |
| | Vicat softening temperature | PASS |
| | Charpy Impact strength | PASS |
| | Flexural modulus of elasticity | PASS |
| | Tensile impact strength at 23 C | PASS |
| 5.2 | Appearance | PASS |
| 5.3 | Dimensions and tolerances | |
| 5.3.2 | Thickness of walls of main profile | PASS |
| 5.3.3 | Tolerances on other dimensions | PASS |
| 5.3.4 | Deviation from straightness of main profiles | PASS |
| 5.4 | Mass of main profiles | PASS |
| 5.5 | Heat reversion | PASS |
| 5.6 | Resistance to impact of main profiles by falling mass | PASS |
| 5.7 | Behaviour after heating at 150 C | PASS |
| 5.8 | Weldability | PASS |

# Glossary of Terms.

PASS: Complies. Tested by BSI engineers at BSI laboratories.

PASS1: Complies. Witnessed by BSI engineers in manufacturers laboratory.

PASS2: Complies. Tests carried out by third party lab; results accepted by BSI.

PASS*: Report resulted in uncertainty and states that Compliance is more probable than non-compliance.

FAIL: Non compliance – Product does not meet the requirements of this clause.

FAIL*: Report resulted in uncertainty and states that Non-compliance is more probable than compliance.

N/A: Not applicable to design under consideration.

N/T: Not tested due to similarity to previously tested item; reference earlier test report.

# Conditions of Issue.

This Test Report is issued subject to the conditions stated in current issue of 'BSI Terms of Service'. The results contained herein apply only to the particular sample(s) tested and to the specific tests carried out, as detailed in this Test Report. The issuing of this Test Report does not indicate any measure of Approval, Certification, Supervision, Control or Surveillance by BSI of any product. No extract, abridgement or abstraction from a Test Report may be published or used to advertise a product without the written consent of BSI, who reserve the absolute right to agree or reject all or any of the details of any items or publicity for which consent may be sought.

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BSI
Kitemark House
Maylands Avenue
Hemel Hempstead
Hertfordshire
HP2 4SQ

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# BS EN 12608 Direct

## Test Results.

**CLAUSE**

**5. REQUIREMENTS**

**5.1 Materials**

**5.1.3 Physical Properties**

**Material characteristics**

| | Specified | Actual | Assessment |
|---|---|---|---|
| **Vicat softening temperature** | | | |
| Annex A.4.1 (BS EN ISO 306:1997) | | | |
| Vicat softening temperature (ºC) | 75 min | 80.6 | Pass |
| Samples taken from profile PC 62-US-04 | | | |
| **Charpy Impact strength** | | | |
| Annex A.4.2 (See BS EN ISO 179-2:1999) | | | |
| Arithmetic mean (kJ/m$^2$) | 20 min | 20.94 | Pass |
| Standard deviation (kJ/m$^2$) | - | 9.4 | |
| Co-efficient of variation (%) | - | 45.37 | |
| Samples taken from profile PC 62-US-04 | | | |
| Mean for 6 brittle (P) fractures | | | |
| Arithmetic mean (kJ/m$^2$) | 20 min | 74.50 | Pass |
| Standard deviation (kJ/m$^2$) | - | 2.58 | |
| Co-efficient of variation (%) | - | 3.46 | |
| Samples taken from profile PC 62-US-04 | | | |
| Mean for 4 ductile (P) fractures | | | |
| **Flexural modulus of elasticity** | | | |
| Annex A.4.3 (BS EN ISO 178:2003) | | | |
| Mean flexural modulus of elasticity (MPa) | 2200 min | 2967.5 | Pass |
| Samples taken from profile PC 62-US-04 | | | |
| **Tensile impact strength at 23ºC** | | | |
| Annex A.4.4 (BS EN ISO 8256:1997) | | | |
| Arithmetic mean (kJ/m$^2$) | 600 min | 664.30 | Pass |
| Standard deviation (kJ/m$^2$) | - | 74.85 | |
| Co-efficient of variation (%) | - | 11.27 | |
| Samples taken from profile PC 62-US-04 | | | |



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# Test Results (Continued).

**CLAUSE**

**5. REQUIREMENTS**

**5.2 Appearance**

The colour of the profiles shall be the same and uniform when viewed by normal vision or corrected vision at a range of 1m, in 45º north sky light viewing perpendicular to the surface in accordance with EN ISO 105-A01:1995 or with an equivalent artificial source of light. The surfaces of the profiles shall be smooth, flat and free from pitting, impurities, cavities and other surface defects. The edges of the profiles shall be clean and burr-free.

| Profile code | Assessment |
|---|---|
| PC 62-US-04 | Pass |

# Test Results (Continued).

**CLAUSE**

**5. REQUIREMENTS (CONTINUED)**

**5.3 Dimensions and tolerances**

**5.3.1 Nominal shape**

The cross-section of the profiles shall conform to the nominal profile. The tolerances of the external dimensions of the profile with respect to the nominal profile shape shall be in accordance with Table 4

**5.3.2 Thickness of walls of main profile**

The thickness of the walls of the main profile according to Figure 2 shall be declared by the manufacturer

**5.3.3 Tolerances on other dimensions**

The critical dimensions of main profiles other than the thickness of the external walls and of auxiliary profiles as well as their tolerances shall be specified by the manufacturer

**Profile code:** PC 62-US-04

Class A
See drawing on page 14

Manufacturer's drawing number: None
Issue date: None

| Dimension | Actual measurement (mm) | Assessment |
|---|---|---|
| A | 61.99 | Pass |
| B | 81.84 | Pass |
| C | 82.03 | Pass |
| D | 101.51 | Pass |
| E | 3.01 | Pass |
| F | 2.53 | Pass |

# Test Results (Continued).

**CLAUSE**

**5. REQUIREMENTS (CONTINUED)**

**5.3 Dimensions and tolerances (Continued)**

**5.3.4 Deviation from straightness of main profiles**

When measured in accordance with the method described in Clause 6.2 the deviation from the straightness shall not be greater than 1mm for a length of 1m

| Profile code | Sight surface | Specified | Actual | Assessment |
|---|---|---|---|---|
| PC 62-US-04 | 1 | 1 max | 0.13 | Pass |
| | 2 | 1 max | 0.29 | Pass |

# Test Results (Continued).

**CLAUSE**

**5. REQUIREMENTS (CONTINUED)**

**5.4 Mass of main profiles**

When measured in accordance with the method described in Clause 6.3 the mass per metre length of main profiles shall not be less than 95% of the nominal mass per metre length

| Profile code | Specified (%) | Actual (%) | Assessment |
|---|---|---|---|
| PC 62-US-04 | 95 min | 100.6 | Pass |

# Test Results (Continued).

**CLAUSE**

**5. REQUIREMENTS (CONTINUED)**

**5.5 Heat reversion**

**5.5.1 Main profile**

When tested in accordance with BS EN 479:1999 for each test specimen the heat reversion of the two largest opposing sight surfaces shall not be greater than 2.0%

The difference in heat reversion for each test specimen between these sight surfaces shall not be greater than 0.4%

**5.5.2 Auxiliary profiles**

When tested in accordance with BS EN 479:1999 the heat reversion for each test specimen shall not be greater than 3.0%, for glazing beads used externally, a limit of 2% max is recommended

**Mean reversion**

| Profile code | Actual value | Assessment |
|---|---|---|
| PC 62-US-04 | 0.99 | Pass |
| | 1.01 | Pass |
| | 0.99 | Pass |

**Variation between sight surfaces**

| Profile code | Sample code | Actual value | |
|---|---|---|---|
| PC 62-US-04 | 1 | 0.17 | Pass |
| | 2 | 0.02 | Pass |
| | 3 | 0.05 | Pass |

# Test Results (Continued).

**CLAUSE**

**5. REQUIREMENTS (CONTINUED)**

**5.6 Resistance to impact of main profiles by falling mass**

When main profiles are tested in accordance with BS EN 477:1999 for the appropriate classification no more than one test specimen shall show rupture in the wall

For coextruded profiles the delamination of the coextruded layer is also considered as failure

| Profile Codes | Class | Mass used (g) | Drop height (m) | Number failures out of 10 |
|---|---|---|---|---|
| PC 62-US-04 | 11 | 1000 | 1.5 | 0 |

# Test Results (Continued).

**CLAUSE**

**5. REQUIREMENTS (CONTINUED)**

**5.7 Behaviour after heating at 150°C**

When tested in accordance with BS EN 478:1999 the profiles shall show no defects. For coextruded profiles the delamination of the coextruded layer is also considered as failure

**Profile code**
PC 62-US-04

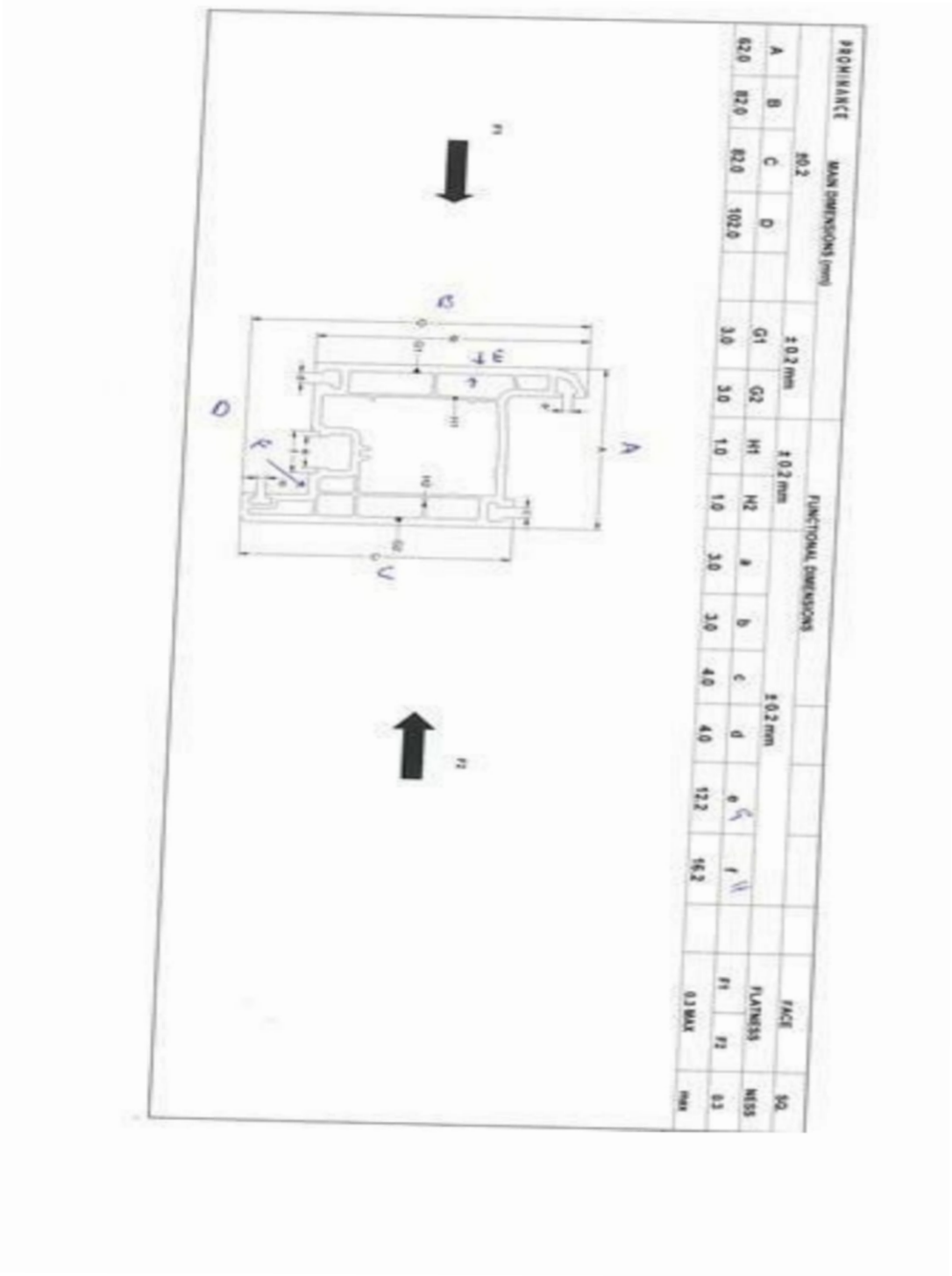
**5.8 Weldability**

When tested in accordance with EN 514:2000, the mean weld failure stress shall not be less than 25 MPa and no individual result shall fall below 20 MPa

| Profile code | I ($mm^4$) | V (mm) | Sample No | Stress (MPa) | Assessment |
|---|---|---|---|---|---|
| PC 62-US-04 | 472276.44 | 31.7 | 1 | 24.34 | Pass |
| | | | 2 | 25.27 | Pass |
| | | | 3 | 26.36 | Pass |
| | | | Mean | 25.32 | Pass |

Note 'I' and 'V' values supplied by manufacturer.

## Manufacturers Drawings.



## Photograph of Samples.

***End of Report***